SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Technical Presentation

Code of Conduct:

I Kishore S.P (192511081) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Technical Presentation

1. **Reactive Agent-Based Traffic Control System**
   Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

Working:

- Collects real-time data from cameras, loop sensors, GPS, and IoT devices.
- Streaming data is processed to determine vehicle density, queue length, and traffic speed.
- If congestion is detected, the agent immediately changes signal timings.
- Gives longer green time to heavily congested directions.
- Reduces green time for low-traffic directions.
- Continuously repeats the process as new data arrives.

Pipeline:

Goal: Minimize waiting time, queue length, and congestion.

2. **Goal-Based Agent for Healthcare Diagnosis**
   Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

Working:

- Collects patient information such as symptoms, medical history, laboratory results, and imaging data.
- Defines a diagnostic goal.
- Compares the available patient evidence with possible diseases.
- Evaluates possible diagnostic and treatment actions.
- Selects the action that best satisfies the goal.
- Updates the decision when new patient data becomes available.

Pipeline:

The system should support clinicians rather than replace professional medical judgment.

3. **Utility-Based Agent for E-Commerce Recommendations**
   Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

A **Utility-Based Agent** recommends products by assigning a utility or preference score to different products.

The utility can depend on:

- User preferences
- Previous purchases
- Product price
- Ratings
- Brand preference
- Purchase history
- Current browsing behavior

For example:
The agent calculates utility scores and recommends products with the highest expected utility.
Pipeline:
Large-scale data processing can be used to process millions of users and products efficiently.

4. Learning Agent for Personalized Education System
Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

A **Learning Agent** improves its behavior based on feedback from students.
Working:
- Monitors student performance, quiz results, time spent, and mistakes.
- Identifies strengths and weaknesses.
- Adjusts the difficulty of learning materials.
- Recommends additional exercises when a student performs poorly.
- Increases difficulty when the student consistently performs well.
- Learns from continuous feedback.

PySpark can process large amounts of student activity data in distributed systems.
Pipeline:
Goal: Provide adaptive learning and improve student performance over time.

5. Multi-Agent System for Disaster Management
Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

A **Multi-Agent System (MAS)** uses multiple specialized agents that cooperate to detect and respond to disasters.
Sensor Agents
Collect data from:
- Cameras
- Weather sensors
- Seismic sensors
- Drones
- IoT devices

Decision Agents
Analyze sensor information, estimate disaster severity, and determine appropriate actions.
Communication Agents
Share alerts and instructions with:
- Emergency services
- Hospitals
- Rescue teams
- Government authorities
- The public

Coordination
Agents exchange information and coordinate their actions to avoid conflicts and respond quickly.

Pipeline:
Advantages:

- Distributed decision-making
- Faster response
- Fault tolerance
- Real-time information sharing
- Ability to handle large-scale disasters

Example: During a flood, sensor agents detect rising water levels, decision agents determine the affected areas, and communication agents send evacuation alerts to residents and rescue teams.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach

		advanced methods			
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness